

| Sensor Device                           |         |                                                                                                                                                                                                                                     |       |           |                       |
|-----------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|-----------------------|
| System Parameters (defined by hardware) |         | Profiles (usage of each component mode - defined by software)                                                                                                                                                                       |       |           |                       |
| form the datasheets                     |         | datasheet                                                                                                                                                                                                                           | "off" | "sensing" | "interactive"         |
| <b>Processor</b>                        |         | <a href="https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/ESP32-C3-DevKitC-1-User-Guide.pdf">https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/ESP32-C3-DevKitC-1-User-Guide.pdf</a> |       |           |                       |
| Active                                  | 85 mW   |                                                                                                                                                                                                                                     | 0%    | 20%       | 50%                   |
| Idle                                    | 33 mW   |                                                                                                                                                                                                                                     | 0%    | 10%       | 50%                   |
| Sleep                                   | 0 mW    |                                                                                                                                                                                                                                     | 100%  | 70%       | 0%                    |
| <b>LED</b>                              |         |                                                                                                                                                                                                                                     |       |           |                       |
| On                                      | 7 mW    |                                                                                                                                                                                                                                     | 0%    | 5%        | 30%                   |
| <b>Sensor</b>                           |         | <a href="https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/soilMoistureSensorV1.0.pdf">https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/soilMoistureSensorV1.0.pdf</a>               |       |           |                       |
| On                                      | 17 mW   | <a href="https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/soilMoistureSensorV1.0.pdf">https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/soilMoistureSensorV1.0.pdf</a>               | 0%    | 20%       | 30%                   |
| Idle                                    | 17 mW   | <a href="https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/soilMoistureSensorV1.0.pdf">https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/soilMoistureSensorV1.0.pdf</a>               | 100%  | 80%       | 70%                   |
| Off                                     | 0 mW    |                                                                                                                                                                                                                                     | 0%    | 0%        | 0%                    |
| <b>Display</b>                          |         |                                                                                                                                                                                                                                     |       |           |                       |
| On                                      | 0 mW    |                                                                                                                                                                                                                                     | 0%    | 0%        | 0%                    |
| Off (leakage)                           | 0 mW    |                                                                                                                                                                                                                                     | 0%    | 0%        | 0%                    |
| <b>Radio</b>                            |         | <a href="https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/ESP32-C3-DevKitC-1-User-Guide.pdf">https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/ESP32-C3-DevKitC-1-User-Guide.pdf</a> |       |           |                       |
| Data Rate                               | 300 bps |                                                                                                                                                                                                                                     | 0%    | 0%        | 0%                    |
| Standby Power                           | 5 mW    |                                                                                                                                                                                                                                     | 0%    | 95%       | 70%                   |
| TX Power                                | 20 mW   |                                                                                                                                                                                                                                     | 0%    | 3%        | 20%                   |
| RX Power                                | 10 mW   |                                                                                                                                                                                                                                     | 0%    | 2%        | 10%                   |
| <b>Battery</b>                          |         |                                                                                                                                                                                                                                     |       |           |                       |
| Capacity                                | 500 mAh |                                                                                                                                                                                                                                     | 22    | 1.5       | 0.5 hours/day typical |
| Nominal Voltage                         | 3 V     |                                                                                                                                                                                                                                     |       |           |                       |
| Regulator Efficiency                    | 95%     |                                                                                                                                                                                                                                     |       |           |                       |

supply. Power values

REFLECTIONS : WHAT DID YOU LEARN FROM ANALYZING YOUR POWER. TALK ABOUT SOME POTENTIAL TRADEOFFS. that WiFi (TX/RX

ware and usage)

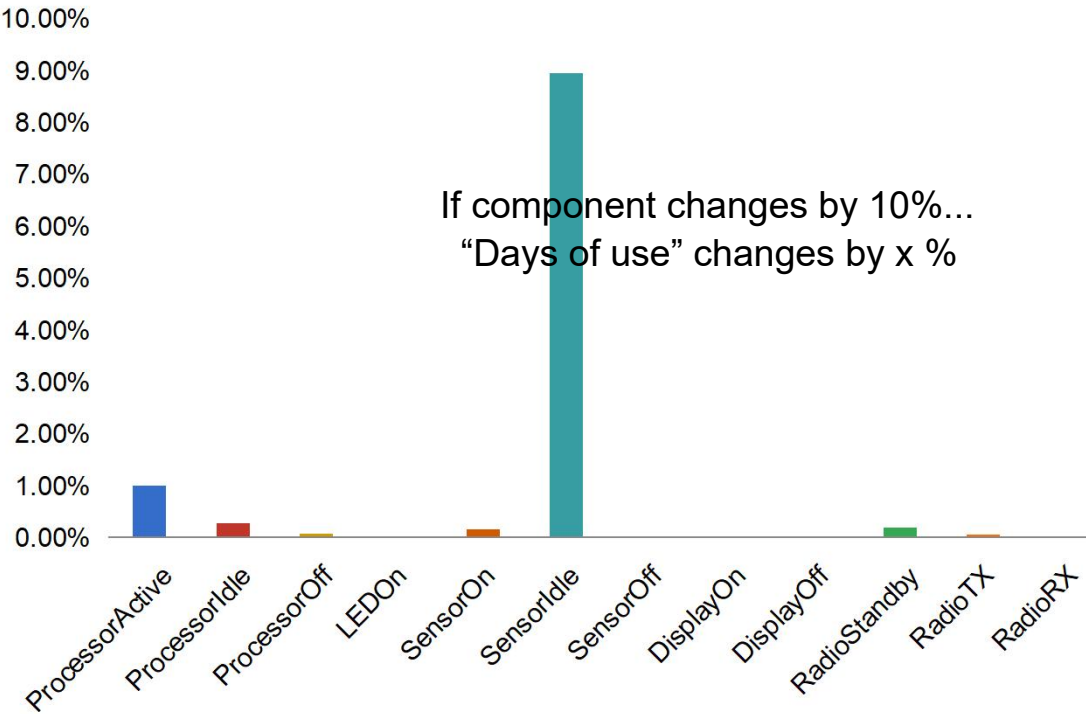
32.pdf

%20moisture%20sensor.pdf

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al usage

# Sensitivity Analysis



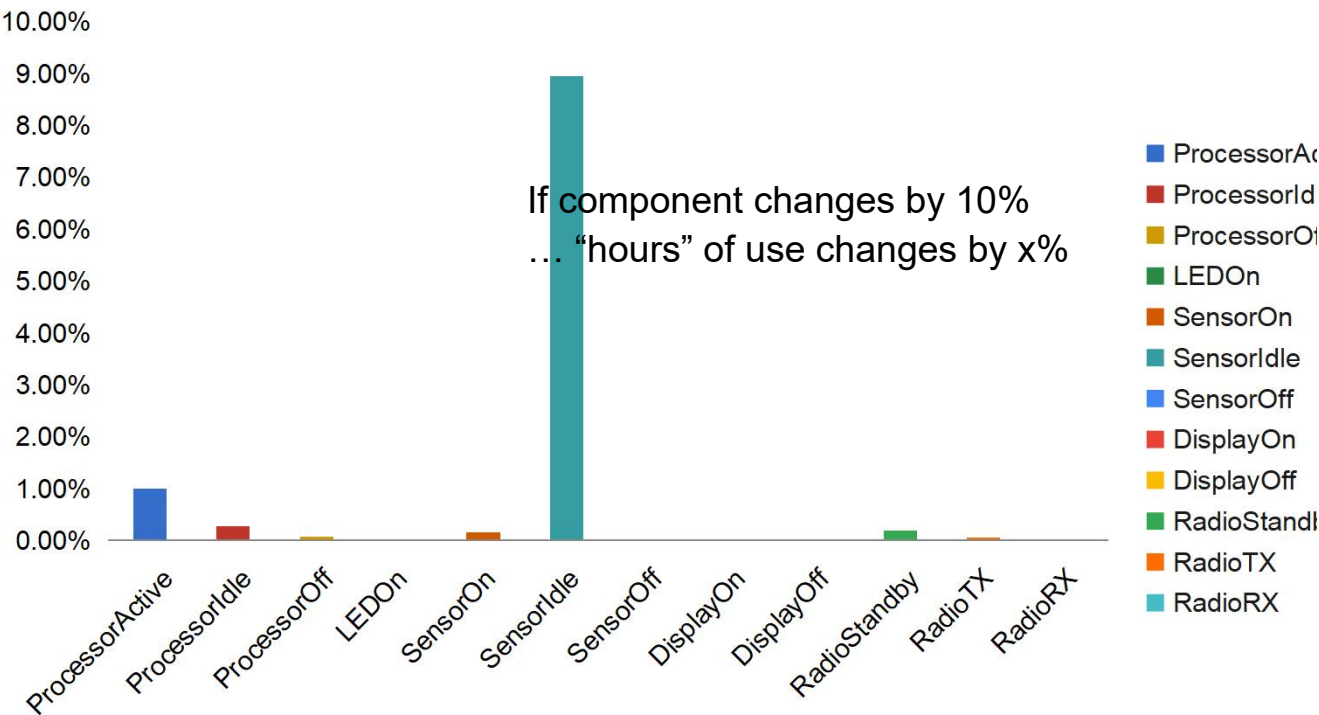
| Total power in profile (mw) |              | Maximum Time |
|-----------------------------|--------------|--------------|
| "off"                       | 16.645 mW    | 94.2 hours   |
| "sensing"                   | 42.823519 mW | 36.6 hours   |
| "interactive"               | 86.005014 mW | 18.2 hours   |

Effective Battery Capacity  
1567.5 mW\*h

|              |             |
|--------------|-------------|
| Days of Use  | 3.31 days   |
| Hours of Use | 79.46 hours |

- ProcessorActive
- ProcessorIdle
- ProcessorOff
- LEDOn
- SensorOn
- SensorIdle
- SensorOff
- DisplayOn
- DisplayOff
- RadioStandby
- RadioTX
- RadioRX

# Sensitivity Analysis



Linked Data from First Page DO NOT EDIT

|                      |       |   |                                                                              |           |    |    |
|----------------------|-------|---|------------------------------------------------------------------------------|-----------|----|----|
| System Parameter     | 0     | 0 | 0                                                                            |           |    |    |
| 0 from the datasheet |       | 0 | 0                                                                            |           |    |    |
| 0                    | 0     | 0 | 0                                                                            | datasheet |    |    |
| Processor Active     | 0     | 0 | https://github.com/Sienna-Zhang/514_final/blob/main/514_final/datasheets/ESF |           |    |    |
|                      | 85 mW |   | 76.329                                                                       | 85        | 85 | 85 |

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|   |   |
|---|---|
| 0 | 0 |
| 0 | 0 |
| 0 | 0 |
| 0 | 0 |
| 0 | 0 |
| 0 | 0 |

|                 |           |               |   |
|-----------------|-----------|---------------|---|
| Profiles (usage | 0         | 0             | 0 |
| 0               | 0         | 0             | 0 |
| "off"           | "sensing" | "interactive" | 0 |
| 0               | 0         | 0             | 0 |
| 0%              | 20%       | 50%           | 0 |

[illegible]

[illegible]